**4** Fig. 4.1 shows a bottle part-filled with water. The air inside the bottle is at the same pressure as the air outside the bottle. The bottle and its contents are at room temperature.

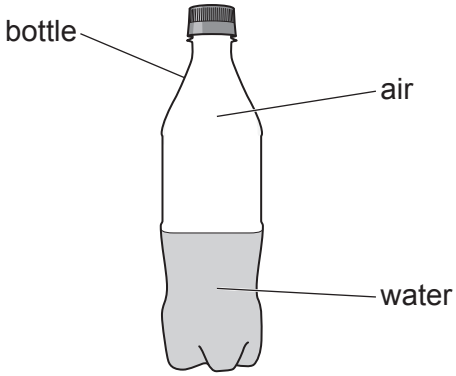



**Fig. 4.1**

**(a)** The temperature of the bottle and its contents are increased.

**(i)** Explain, in terms of particles, how the air pressure inside the bottle changes as the temperature increases.

.....

.....

.....

..... [3]

**(ii)** The lid is removed from the bottle.

State and explain how the air pressure inside the bottle changes.

statement .....

explanation .....

.....

[2]

**(b)** The mass of water in the bottle is 0.18 kg. The specific heat capacity of water is 4200 J / (kg °C).

Calculate the thermal energy needed to increase the temperature of the water by 20 °C.

thermal energy = ..... [2]

**(c)** Another plastic bottle is filled to the top with water. The height of the bottle is 40.0 cm. The density of water is $1.0 \times 10^3$ kg / m$^3$.

Calculate the pressure difference between the top and bottom of the water.

pressure difference = ..... [2]

[Total: 9]